

Deliverable1 - General information

We are going to have two Deliverables

- D1 will focus on the **first part of the course**
- D2 will be related to **shared memory** optimizations **and other advanced features.**

Grade balancing:

- D1: **40 %** (deadline: **May 1-7 2026 at 23.59**)
- Final Deliverable: **60 %** (deadline exam session)

Each deliverable will be assessed based on:

- Code: 30% of the grade
- Report: 70% of the grade

Check the first lectures and ask for any clarification

Deliverable1 - Problem statement

- Write a simple algorithm that performs a **Sparse-Matrix dense-Vector multiplication** (SpMV).
- The algorithm takes the input file path as input and generates a random vector. Example: `“./SpMV path/to/input.mtx”`
- Input files are available in the **Suitsparse dataset** (<https://sparse.tamu.edu/>)
- For the analysis:
 - Write a CPU implementation as a baseline and different GPU implementations which use different optimization strategies.
 - Measure the performance and benchmark your implementation.
 - Profile your implementations and analyze your bottlenecks.

Deliverable 1 - input dataset

Supplementary exercise:

- Download the matrix marker file from Suitsparse (<https://sparse.tamu.edu/>) and read the sparse matrix from the file.

Parse mtx example:

https://math.nist.gov/MatrixMarket/mmio/c/example_read.c

Displaying collection matrices 1 - 20 of 2893 in total

← 1 2 3 4 5 6 7 8 9 ... 144 145 →

Id	Name	Group	Rows	Cols	Nonzeros	Kind	Date	Download File
1	1138_bus	HB	1,138	1,138	4,054	Power Network Problem	1985	MATLAB Rutherford Boeing Matrix Market

```
[lorenzo.pichetti@marzola ~]$ wget https://suitsparse-collection-website.herokuapp.com/MM/HB/1138_bus.tar.gz
[...]  
[lorenzo.pichetti@marzola ~]$ gzip -d 1138_bus.tar.gz  
[lorenzo.pichetti@marzola ~]$ tar -xf 1138_bus.tar  
[lorenzo.pichetti@marzola ~]$ ls 1138_bus/1138_bus.mtx  
1138_bus/1138_bus.mtx
```

Deliverable1 - Report structure

Send a report (up to 4 pages) with:

- Your name – student ID – email address
- Link to git repository

Structure:

- **Introduction** – problem description, limits, proposals
- **Methodology** - description of the algorithm, pseudo-code,
- **Experiments:**
 - Hardware and software / library used
 - Plots describing the performance and the variation of the performance.
(e.g., x-axis matrix dimensions, y-axis effective-bandwidth)
- **References**